

Multi-Agent Scientific Paper Review

qwen3:8b · May 20, 2026 16:12

File	REVISTA_Project.pdf
Words	6,252
Sections Detected	preamble, introduction, methodology, experiments, results, discussion, conclusions, references, abstract,
Review Mode	DEEP
Model	qwen3:8b
Target Publisher	IEEE

■ **AI-ASSISTED REVIEW NOTICE:** This report was generated by CHVN Paper Reviewer, an AI-powered multi-agent system running locally. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.

Editorial Decision

MAJOR REVISION**Weighted Score: 3.2 / 5.0**
Confidence: 84%

Executive Summary

The manuscript presents a novel engineering design for solar-powered refrigeration aimed at food preservation in rural areas. While the paper provides detailed technical descriptions and quantitative results, it lacks sufficient experimental validation, reproducibility, and comprehensive literature review.

Decision Rationale

The manuscript has significant technical merit but requires major revisions to address critical issues related to reproducibility, experimental validation, and comprehensive literature review. The lack of proper experimental procedures, insufficient comparison with state-of-the-art systems, and incomplete references prevent the manuscript from meeting IEEE standards for publication.

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.5	Excellent	■■■■■
Manuscript Structure	2.5	Weak	■■■■■
Methodology & Equations	3.5	Good	■■■■■
Statistical Analysis	4.0	Good	■■■■■
Results & Evidence	3.5	Good	■■■■■

Discussion & Conclusions	2.5	Weak	■■■■■
Figures & Tables	4.5	Excellent	■■■■■
Scientific Writing	2.5	Weak	■■■■■
References & Citations	1.5	Very Weak	■■■■■
Ethics & Transparency	1.2	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.5 / 5.0 | Confidence: 95%

Strengths:

- [TITLE] Clearly states the purpose and scope of the work
- [ABSTRACT] Follows structured format with clear background, objective, methods, results, and conclusions
- [ABSTRACT] Quantifies results with specific percentages and performance metrics
- [KEYWORDS] Includes domain-specific terms and relevant indexing terms for discoverability

Weaknesses:

- [TITLE] Could be more concise while retaining specificity
- [KEYWORDS] Contains title duplication and may exceed recommended keyword count

Major Comments:

- [KEYWORDS] Avoid duplication of title terms and reduce keyword count to 4-8 for better indexing efficiency

Minor Comments:

- [TITLE] Could be shortened slightly for improved readability without losing specificity

Recommendations:

- [TITLE] Consider shortening the title to 15-20 words for better scanning and discoverability
- [KEYWORDS] Revise the keyword list to remove title duplicates and ensure alignment with the content and IEEE indexing standards

StructureReviewer

Manuscript structure and logical coherence

Score: 2.5 / 5.0 | Confidence: 95%

Strengths:

- Title and abstract clearly state the problem of food preservation in rural areas using solar energy.

Weaknesses:

- Introduction lacks a clear statement of the research problem and hypothesis.
- Research objectives are not explicitly stated and are not fully addressed in the conclusions.
- Methodology section is incomplete and lacks detailed description of the experimental setup and procedures.
- Experiments section is fragmented and lacks a coherent structure.
- Results section is incomplete and lacks quantitative data to support claims.
- Discussion section contains equations without proper numbering, definitions, or dimensional consistency.
- Conclusions do not clearly align with the stated objectives or results.
- References are incomplete, with missing citations and some entries that are not properly formatted.
- Keywords are missing and not aligned with the content.
- Abstract lacks specific details about the methodology, results, or conclusions.

Major Comments:

- The manuscript fails to clearly state the research problem, hypothesis, or objectives in the introduction. This undermines the logical flow and coherence of the entire document.
- The methodology section is incomplete and lacks sufficient detail to allow for reproducibility. The experimental procedures are not clearly described, making it impossible to replicate the work.
- The results section is incomplete and lacks quantitative data to support the claims made in the discussion and conclusions.

- The discussion section contains equations that are not properly numbered, defined, or dimensionally consistent. For example, the equation for COPCarnot is presented without proper formatting or explanation.
- The conclusions do not clearly align with the stated objectives or results. The manuscript fails to demonstrate how the findings address the research problem or hypothesis.
- The references section is incomplete, with missing citations and some entries that are not properly formatted. This affects the credibility and academic rigor of the manuscript.
- The abstract lacks specific details about the methodology, results, or conclusions, making it difficult to assess the contribution of the work.

Minor Comments:

- The title is incomplete and lacks a clear statement of the research focus.
- Keywords are missing and should be included to improve searchability and alignment with the content.
- The experiments section is fragmented and lacks a coherent structure, making it difficult to follow the experimental process.

Recommendations:

- Revise the introduction to clearly state the research problem, hypothesis, and objectives.
- Expand the methodology section to include detailed descriptions of the experimental setup, procedures, and data collection methods.
- Complete the results section with quantitative data and statistical analysis to support the claims made in the discussion and conclusions.
- Ensure that all equations are properly numbered, defined, and dimensionally consistent.
- Revise the conclusions to clearly align with the stated objectives and results, demonstrating how the findings address the research problem.
- Complete the references section with properly formatted citations and ensure all references are relevant and recent.
- Include keywords that accurately reflect the content of the manuscript.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear problem statement and justification for the proposed solution.
- Detailed description of the design and construction process.
- Thermodynamic analysis and comparison with theoretical limits.
- Use of IoT for real-time monitoring and data collection.
- Clear alignment between objectives and conclusions.

Weaknesses:

- The 'Experiments' section is incomplete and lacks detailed experimental procedures.
- The 'Results' section is fragmented and does not clearly link to the experimental setup described.
- The 'Discussion' section references results that are not fully presented in the 'Results' section.
- Equations (6) and (7) are not fully defined with all variables explicitly stated.
- The paper lacks a detailed description of the experimental setup and data collection procedures.

Major Comments:

- The 'Experiments' section is incomplete and lacks detailed experimental procedures. The paper should provide a clear description of the experimental setup, data collection methods, and testing procedures to ensure reproducibility.
- The 'Results' section is fragmented and does not clearly link to the experimental setup described. The results should be presented in a structured manner that aligns with the experimental procedures.

- The 'Discussion' section references results that are not fully presented in the 'Results' section. The results should be fully presented before discussing their implications.
- Equations (6) and (7) are not fully defined with all variables explicitly stated. The paper should define all variables used in the equations to ensure clarity and reproducibility.

Minor Comments:

- The 'Experiments' section should be expanded to include detailed experimental procedures and data collection methods.
- The 'Results' section should be structured to clearly link to the experimental setup described.
- The 'Discussion' section should reference results that are fully presented in the 'Results' section.
- Equations (6) and (7) should be fully defined with all variables explicitly stated.

Recommendations:

- Expand the 'Experiments' section to include detailed experimental procedures and data collection methods.
- Structure the 'Results' section to clearly link to the experimental setup described.
- Ensure that all results are fully presented before discussing their implications in the 'Discussion' section.
- Define all variables used in equations (6) and (7) explicitly.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **4.0 / 5.0** | Confidence: 75%

Strengths:

- Engineering design paper: inferential statistical tests are not required for this paper type.
- Measurement uncertainty or variability is acknowledged in the manuscript.
- Operating conditions are described for the reported performance metrics.
- Measurement instruments or data collection methods are mentioned.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.5 / 5.0** | Confidence: 95%

Strengths:

- All figures are self-explanatory and referenced in the text before they appear.
- Tables are clearly labeled with headers and units, and their data is consistent with the text.
- Equations are numbered sequentially and defined in context, with dimensional consistency.
- The manuscript provides a clear and reproducible methodology for the design and testing of the refrigeration system.
- The text and figures align well, supporting the claims made about the system's performance and efficiency.

Weaknesses:

- Some figures (e.g., Fig. 6) are described in the text but not fully detailed in the manuscript, which may limit reproducibility.
- The text mentions the use of IoT for monitoring but does not provide detailed information on the data acquisition frequency or sampling intervals.
- The COP calculation in Eq. (6) is presented without a clear derivation or reference to the thermodynamic model used.
- The text references a 'Fig. 6' but does not provide the full figure or its content, which may affect the reproducibility of the results.
- The text discusses the use of the DHT11 sensor for temperature and humidity but does not specify the sampling rate or calibration procedures.

Major Comments:

- The manuscript should provide the full content of all figures, especially Fig. 6, to ensure reproducibility and clarity for independent researchers.
- The derivation of the COP calculation in Eq. (6) should be included in the methodology section to ensure mathematical rigor and transparency.
- The text should clarify the sampling frequency and calibration procedures for the DHT11 and SHT40 sensors to ensure data reliability.

Minor Comments:

- The text could benefit from more precise descriptions of the IoT system's architecture and data transmission protocols.
- The table on refrigerant properties (Table 2) should include units for all values to ensure clarity and consistency.
- The text could clarify the exact conditions used for the thermodynamic analysis (e.g., pressure, temperature, and flow rates) to ensure reproducibility.

Recommendations:

- Include the full content of all figures, especially Fig. 6, to ensure reproducibility and clarity.
- Provide a detailed derivation of the COP calculation in Eq. (6) in the methodology section.
- Clarify the sampling frequency and calibration procedures for the DHT11 and SHT40 sensors.
- Add units to all values in Table 2 for consistency and clarity.
- Specify the exact conditions used for the thermodynamic analysis (e.g., pressure, temperature, and flow rates) to ensure reproducibility.

ResultsReviewer

Results validity and empirical support

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- The manuscript provides detailed quantitative results such as COP, heat removal rates, and energy consumption, which are essential for evaluating the system's performance.
- The comparison of R-290 and R-134a is well-supported with a table that includes relevant properties and performance metrics.
- The system design and implementation are described with sufficient detail for reproducibility, including the IoT monitoring system and its components.
- The volume and temperature range of the prototype are clearly defined and aligned with the stated objectives.

Weaknesses:

- The results section lacks a clear distinction between the prototype's performance and the baseline or state-of-the-art systems, which is critical for establishing novelty and impact.
- The claim about the COP being 75% of the Carnot COP is not substantiated with a calculation or reference to the Carnot COP under the same conditions.
- The results section does not clearly state whether the prototype has been fully constructed and tested, which is essential for validating the claims made in the abstract and introduction.
- The results are not consistently presented across text, figures, and tables. For example, the COP value is mentioned in the abstract and results but not clearly tied to a specific figure or table.
- The results section does not include statistical analysis or confidence intervals for the reported performance metrics, which is necessary for assessing the reliability of the findings.

Major Comments:

- The claim that the COP of 4.72 represents 75% of the ideal Carnot COP is not supported by any calculation or reference to the Carnot COP under the same conditions. This is a critical claim that needs to be substantiated with a proper calculation.

- The results section lacks a clear comparison with baseline or state-of-the-art systems, which is essential for demonstrating the novelty and impact of the proposed solution.
- The manuscript does not clearly state whether the prototype has been fully constructed and tested, which is necessary for validating the claims made in the abstract and introduction.

Minor Comments:

- The results section does not consistently present data across text, figures, and tables. For example, the COP value is mentioned in the abstract and results but not clearly tied to a specific figure or table.
- The results section does not include statistical analysis or confidence intervals for the reported performance metrics, which is necessary for assessing the reliability of the findings.

Recommendations:

- Include a calculation or reference to the Carnot COP under the same operating conditions to substantiate the claim that the COP of 4.72 represents 75% of the ideal Carnot COP.
- Provide a clear comparison with baseline or state-of-the-art systems to demonstrate the novelty and impact of the proposed solution.
- Clarify whether the prototype has been fully constructed and tested, and include the results of any testing or validation.
- Ensure that all quantitative results are consistently presented across text, figures, and tables, and include statistical analysis or confidence intervals for the reported performance metrics.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical calculations and equations with proper numbering and definitions
- Reproducibility of the methodology is implied through detailed steps
- Comparison of COP values with prior systems (absorption solar, thermoelectric) is provided

Weaknesses:

- Discussion lacks a comprehensive comparison with prior work beyond COP values
- Discussion does not address limitations of the study
- Discussion overgeneralizes the applicability of the system to rural areas without specific data
- Conclusions section is incomplete and lacks specific quantified outcomes
- No clear statement of contribution to the field or theoretical implications

Major Comments:

- The Discussion section fails to address the limitations of the study, which is a critical component of a scientific manuscript.
- The Discussion does not provide a thorough comparison with prior work, only citing COP values without detailed analysis.
- The Discussion overgeneralizes the applicability of the system without supporting data or context.
- The Conclusions section is incomplete and lacks specific, quantified outcomes that directly follow from the results.
- The manuscript does not clearly state the contribution to the field or theoretical implications of the work.

Minor Comments:

- The Discussion section should clarify the assumptions made in the energy consumption estimation.
- The Conclusions section should be expanded to include future work directions and specific recommendations for improvement.

Recommendations:

- Include a detailed comparison with prior work in the Discussion section, not just citation of COP values.

- Address the limitations of the study in the Discussion section.
- Clarify the assumptions and context for the overgeneralization of the system's applicability.
- Expand the Conclusions section to include specific, quantified outcomes and future work directions.
- Provide a clear statement of the contribution to the field and theoretical implications.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical description of the prototype and its components
- Use of specific data and figures to support claims
- Relevant references to prior work and standards

Weaknesses:

- Overuse of passive voice
- Lack of definition for acronyms
- Vague quantifiers without numerical support
- Inconsistent terminology (e.g., 'COP' and 'COP real' used interchangeably without clear distinction)
- Unclear methodology for experimental validation
- Missing statistical analysis and confidence intervals
- Lack of reproducibility details for the prototype construction
- Inconsistent use of units and formatting (e.g., 'COP real' vs. 'COP')

Major Comments:

- The manuscript lacks a clear, reproducible methodology for the experimental validation of the prototype. The description of the prototype construction is insufficient for an independent researcher to replicate the work.
- The use of passive voice is excessive and undermines the clarity and directness of the writing.
- The manuscript contains several undefined acronyms (e.g., DANE, DNP, FAO, IEC, ISO, R290) that should be defined on first use.
- The vague quantifiers such as 'many', 'several', and 'significant' are used without numerical support, which weakens the scientific rigor of the claims.
- The results section lacks statistical analysis, including confidence intervals and effect sizes, which is critical for validating the findings.
- The discussion section contains several unexplained references to figures and equations (e.g., 'Fig. 6', 'Eq. 6') without clear context or explanation.
- The conclusion section is incomplete and does not clearly summarize the key findings or their implications.
- The manuscript contains inconsistencies in terminology, such as the use of 'COP real' and 'COP' interchangeably without clear distinction.

Minor Comments:

- The abstract and introduction should be more aligned in structure and content.
- The keywords section should be more precise and aligned with the content of the manuscript.
- The title should be more specific and include the key technical terms (e.g., 'vapor-compression refrigeration', 'R-290').
- The figure and table captions are not sufficiently descriptive and lack context.
- The use of mixed languages (Spanish and English) is inconsistent and should be standardized.

Recommendations:

- Define all acronyms on first use (e.g., DANE, DNP, FAO, IEC, ISO, R290).

- Revise the methodology section to provide a clear, step-by-step description of the prototype construction and testing procedures.
- Include statistical analysis and confidence intervals in the results section.
- Clarify the distinction between 'COP real' and 'COP' in the discussion and results sections.
- Ensure that all figures and tables have descriptive, self-contained captions that explain their content.
- Revise the conclusion section to clearly summarize the key findings and their implications.
- Standardize the language to either Spanish or English, and avoid mixing the two.
- Revise the abstract and introduction to ensure they are consistent in structure and content.

ReferencesReviewer

References quality, recency, and relevance

Score: **1.5 / 5.0** | Confidence: 70%

Strengths:

- References are recent (2021-2026) and relevant to the topic of rural electrification and food preservation.

Weaknesses:

- The references section is extremely limited in scope and depth, with only two entries.

Major Comments:

- The references section is insufficient for a Q1 journal. It lacks seminal foundational works and does not provide a comprehensive literature review.

Minor Comments:

- The citation style is inconsistent and appears to be a mix of informal and formal formats.

Recommendations:

- Expand the references section to include seminal works in refrigeration, solar energy, and IoT monitoring.
- Ensure that all citations follow a consistent and formal style (e.g., IEEE or APA).
- Include more recent and diverse references to strengthen the literature review.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.2 / 5.0** | Confidence: 95%

Strengths:

- Clear technical focus on a novel engineering design for solar-powered refrigeration.
- Detailed thermodynamic analysis with equations and figures.
- Relevant citations to prior work on solar refrigeration.

Weaknesses:

- Lack of transparency regarding AI tool use, which is a mandatory disclosure for IEEE journals.
- No conflict of interest declaration, which is required for all journals.
- No funding disclosure, which is mandatory for all journals.
- No limitations section, which is essential for responsible reporting.
- No data availability statement, which is critical for reproducibility.
- No clear description of how the prototype was validated or tested, which is essential for engineering papers.
- No mention of how the thermodynamic properties of R-290 were determined, which is necessary for scientific credibility.

Major Comments:

- The manuscript fails to disclose the use of generative AI tools, which is a mandatory requirement under IEEE/COPE 2024 guidelines. This omission raises significant concerns about transparency and compliance.

- There is no conflict of interest declaration, which is required for all journals. This is a major omission that undermines the credibility of the work.
- No funding disclosure is provided, which is mandatory for all journals. This is a critical transparency issue.
- The manuscript lacks a limitations section, which is essential for responsible scientific reporting. The absence of this section prevents readers from understanding the scope and constraints of the work.
- No data availability statement is provided, which is necessary for reproducibility. Without this, independent replication of the work is not possible.
- The manuscript does not clearly describe how the prototype was validated or tested, which is essential for engineering papers. This omission reduces the scientific rigor of the work.
- The thermodynamic properties of R-290 are referenced but not clearly explained, which is necessary for scientific credibility.

Minor Comments:

- The manuscript does not provide a clear description of the experimental setup or validation process for the prototype.
- The equations are not numbered or defined, which is a critical requirement for IEEE publications.
- The figure caption for Fig. 6 is incomplete and lacks sufficient detail for reproducibility.

Recommendations:

- Include a detailed limitations section that explicitly addresses the scope, constraints, and potential generalizability of the work.
- Provide a clear data availability statement and describe the conditions under which the prototype was tested.
- Include a conflict of interest declaration and funding disclosure as required by all journals.
- Disclose the use of any generative AI tools, as required by IEEE/COPE 2024 guidelines.
- Ensure all equations are numbered, defined, and dimensionally consistent.
- Provide sufficient detail in figure captions to allow for independent replication of the work.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [KEYWORDS] Avoid duplication of title terms and reduce keyword count to 4-8 for better indexing efficiency
- [2] The manuscript fails to clearly state the research problem, hypothesis, or objectives in the introduction. This undermines the logical flow and coherence of the entire document.
- [3] The methodology section is incomplete and lacks sufficient detail to allow for reproducibility. The experimental procedures are not clearly described, making it impossible to replicate the work.
- [4] The results section is incomplete and lacks quantitative data to support the claims made in the discussion and conclusions.
- [5] The discussion section contains equations that are not properly numbered, defined, or dimensionally consistent. For example, the equation for COP_{Carnot} is presented without proper formatting or explanation.
- [6] The conclusions do not clearly align with the stated objectives or results. The manuscript fails to demonstrate how the findings address the research problem or hypothesis.
- [7] The references section is incomplete, with missing citations and some entries that are not properly formatted. This affects the credibility and academic rigor of the manuscript.
- [8] The abstract lacks specific details about the methodology, results, or conclusions, making it difficult to assess the contribution of the work.
- [9] The 'Experiments' section is incomplete and lacks detailed experimental procedures. The paper should provide a clear description of the experimental setup, data collection methods, and testing procedures to ensure reproducibility.
- [10] The 'Results' section is fragmented and does not clearly link to the experimental setup described. The results should be presented in a structured manner that aligns with the experimental procedures.
- [11] The 'Discussion' section references results that are not fully presented in the 'Results' section. The results should be fully presented before discussing their implications.
- [12] Equations (6) and (7) are not fully defined with all variables explicitly stated. The paper should define all variables used in the equations to ensure clarity and reproducibility.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [TITLE] Could be shortened slightly for improved readability without losing specificity
- [2] The title is incomplete and lacks a clear statement of the research focus.
- [3] Keywords are missing and should be included to improve searchability and alignment with the content.
- [4] The experiments section is fragmented and lacks a coherent structure, making it difficult to follow the experimental process.
- [5] The 'Experiments' section should be expanded to include detailed experimental procedures and data collection methods.
- [6] The 'Results' section should be structured to clearly link to the experimental setup described.
- [7] The 'Discussion' section should reference results that are fully presented in the 'Results' section.
- [8] Equations (6) and (7) should be fully defined with all variables explicitly stated.
- [9] The text could benefit from more precise descriptions of the IoT system's architecture and data transmission protocols.
- [10] The table on refrigerant properties (Table 2) should include units for all values to ensure clarity and consistency.
- [11] The text could clarify the exact conditions used for the thermodynamic analysis (e.g., pressure, temperature, and flow rates) to ensure reproducibility.
- [12] The results section does not consistently present data across text, figures, and tables. For example, the COP value is mentioned in the abstract and results but not clearly tied to a specific figure or table.

Specific Improvement Recommendations

1. [TITLE] Consider shortening the title to 15-20 words for better scanning and discoverability
2. [KEYWORDS] Revise the keyword list to remove title duplicates and ensure alignment with the content and IEEE indexing standards
3. Revise the introduction to clearly state the research problem, hypothesis, and objectives.
4. Expand the methodology section to include detailed descriptions of the experimental setup, procedures, and data collection methods.
5. Complete the results section with quantitative data and statistical analysis to support the claims made in the discussion and conclusions.
6. Ensure that all equations are properly numbered, defined, and dimensionally consistent.
7. Revise the conclusions to clearly align with the stated objectives and results, demonstrating how the findings address the research problem.
8. Complete the references section with properly formatted citations and ensure all references are relevant and recent.
9. Include keywords that accurately reflect the content of the manuscript.
10. Expand the 'Experiments' section to include detailed experimental procedures and data collection methods.
11. Structure the 'Results' section to clearly link to the experimental setup described.
12. Ensure that all results are fully presented before discussing their implications in the 'Discussion' section.